

Predicting Fraud Risk Through Authentication and Transaction Behavior in Digital Banking

Business Problem

As more people use online and mobile banking for their everyday financial activities, fraud detection has become increasingly important for financial institutions. Digital banking provides convenience, faster transactions, and easier access to banking services, but it also increases the risk of fraudulent activity such as account takeovers, unauthorized transactions, phishing attacks, and suspicious login attempts. Financial institutions must constantly balance customer convenience with strong fraud prevention systems.

The purpose of this project is to analyze how authentication methods, transaction behavior, and digital activity patterns relate to fraudulent banking transactions. The project focuses on identifying whether variables such as repeated login attempts, suspicious IP indicators, transaction velocity scores, device risk scores, and international transaction flags are associated with higher fraud risk. Another objective is to determine whether certain authentication methods appear to provide stronger protection against fraudulent activity.

The overall goal of this project is to better understand fraud-related behavioral patterns and evaluate how machine learning models can help financial institutions improve fraud detection while minimizing disruptions for legitimate customers.

Background/History

Fraud detection has become one of the most important challenges in the banking industry due to the rapid growth of digital banking services. In earlier years, financial institutions primarily relied on rule-based systems that flagged suspicious transactions based on predefined conditions such as unusually large withdrawals, multiple failed login attempts, or transactions occurring from unusual locations. Although these systems provided some protection, fraud techniques have evolved significantly over time, making traditional rule-based methods less effective when used alone.

Today, many financial institutions use machine learning and predictive analytics to improve fraud detection capabilities. Machine learning models can analyze large amounts of transactional and behavioral data in real time to identify suspicious activity patterns that may not be easily detected using fixed rules. These systems help banks reduce fraud losses, improve transaction monitoring, and strengthen cybersecurity efforts.

Authentication systems have also evolved significantly in digital banking environments. Financial institutions now use methods such as multi-factor authentication, one-time passwords (OTP), biometric verification, device fingerprinting, and adaptive authentication systems to improve security. Even with these improvements, fraud remains a major concern because cybercriminals continue to develop new attack methods involving phishing, credential theft, social engineering, and account takeover attempts.

Fraud detection datasets also create challenges for machine learning analysis because fraudulent transactions are typically much less common than legitimate transactions. This imbalance makes it necessary to evaluate machine learning models carefully using metrics beyond simple accuracy.

Data Explanation

The dataset used for this project was obtained from Kaggle and is titled *Banking Fraud Detection & Risk Analytics Dataset*. The dataset contains transactional, behavioral, and authentication-related variables associated with digital banking activity. The dataset includes approximately 10,000 records and multiple features related to customer behavior, transaction activity, security indicators, and fraud outcomes.

Some of the primary variables included in the dataset are:

- authentication type
- login attempts
- suspicious IP indicators
- transaction velocity scores
- device risk scores
- international transaction flags
- session duration
- transaction frequency
- payment channels
- fraud indicators

The target variable for this project is the fraud flag, which identifies whether a transaction was fraudulent or legitimate.

The dataset was cleaned and prepared using Python libraries such as Pandas and NumPy. Data preprocessing included:

- checking for missing values
- validating data consistency
- removing duplicate records
- converting categorical variables into numerical values using encoding methods
- preparing data for machine learning analysis

Exploratory data analysis (EDA) was conducted to better understand fraud distributions, relationships between variables, and behavioral patterns associated with fraudulent transactions.

Methods

This project followed an end-to-end data science workflow that included data preparation, exploratory data analysis, predictive modeling, and business interpretation. Python and Jupyter Notebook were used throughout the project for data cleaning, visualizations, machine learning modeling, and evaluation.

The first stage of the analysis involved exploratory data analysis to identify patterns associated with fraud. Several visualizations were created to better understand fraud distributions and relationships between variables. These visualizations included:

- fraud distribution charts
- fraud rates by authentication type
- correlation heatmaps
- boxplots comparing fraud and non-fraud behavior
- ROC curves
- feature importance charts

Three machine learning classification models were used in this project:

1. Logistic Regression
2. Random Forest Classification
3. XGBoost Classification

Logistic Regression was selected because it provides relatively interpretable results and serves as a strong baseline model. Random Forest and XGBoost were selected because they are more advanced ensemble learning methods that may improve predictive performance.

The models were evaluated using several performance metrics, including:

- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix

Recall was considered especially important because false negatives may allow fraudulent transactions to go undetected.

Analysis

The analysis focused on identifying which behavioral and authentication-related variables were most associated with fraudulent transactions. The exploratory data analysis showed that some variables appeared to have stronger relationships with fraud risk than others. One of the strongest relationships observed in the dataset involved anomaly scores, which showed a noticeably higher average among fraudulent transactions compared to legitimate transactions. Transaction velocity scores also showed slightly higher averages among fraudulent transactions.

However, some variables that were initially expected to show stronger relationships with fraud did not appear to differ significantly between fraud and non-fraud transactions. For example, login attempts showed nearly identical distributions between fraudulent and legitimate transactions. Device risk scores also appeared relatively similar across both groups. These findings demonstrate that not all behavioral variables contribute equally to fraud prediction within this dataset.

The analysis also examined suspicious IP indicators and international transaction activity. While suspicious IP indicators appeared slightly more common among fraudulent transactions, the difference was relatively small. This suggests that suspicious IP activity alone may not strongly differentiate fraud within this dataset and may instead become more useful when combined with additional behavioral indicators.

Another part of the analysis compared fraud rates across different authentication methods to determine whether certain authentication approaches appeared to provide stronger protection

against fraudulent activity. The analysis showed relatively similar fraud rates across authentication methods, although Two-Factor Authentication showed slightly lower fraud rates compared to OTP and Password Only authentication methods.

Correlation analysis and feature importance analysis were also used to better understand relationships between variables. The correlation heatmap showed that anomaly score had the strongest relationship with fraud risk among the variables included in the dataset. Most other variables showed relatively weak individual correlations with fraud. Although correlation does not necessarily prove causation, the analysis suggests that fraud prediction may depend more on combined behavioral patterns rather than a single variable alone.

Three machine learning classification models were trained and evaluated for this project, including Logistic Regression, Random Forest Classification, and XGBoost Classification.

Logistic Regression served as the baseline model for the project. The model provided relatively interpretable results and helped identify which variables had stronger relationships with fraud risk. Although Logistic Regression may not capture highly complex relationships between variables, it still performed reasonably well while maintaining explainability.

The Random Forest model improved predictive performance by using multiple decision trees to capture more complex behavioral patterns. Compared to Logistic Regression, Random Forest reduced prediction errors and improved fraud detection performance.

Feature importance analysis from the Random Forest model showed that variables such as transaction velocity score, device risk score, suspicious IP indicators, and login attempts were among the strongest predictors of fraud.

Among the models tested, XGBoost produced the strongest overall predictive performance. Because XGBoost uses gradient boosting techniques, the model was able to capture more complex interactions between variables and improve fraud prediction accuracy. The ROC-AUC results showed that XGBoost performed better overall compared to Logistic Regression and Random Forest.

Overall, the results suggest that machine learning models can effectively identify fraud-related behavioral patterns in digital banking transactions. More advanced ensemble models such as Random Forest and XGBoost generally performed better than Logistic Regression, although model explainability remains an important consideration for financial institutions.

Conclusion

This project analyzed how authentication methods, transaction behavior, and digital activity patterns relate to fraudulent banking transactions. By applying machine learning techniques to fraud-related behavioral data, the analysis identified several variables associated with higher fraud risk, including suspicious IP activity, transaction velocity scores, repeated login attempts, and device risk scores.

The machine learning models demonstrated that fraud risk can be predicted using behavioral and authentication-related variables. Among the models tested, XGBoost produced the strongest predictive performance, while Logistic Regression provided better interpretability.

The findings from this project may help financial institutions strengthen fraud prevention systems while maintaining a positive customer experience. Understanding which variables are most associated with fraud may also support improvements in authentication systems, behavioral monitoring, and transaction risk analysis.

Assumptions

Several assumptions were made throughout this project.

One assumption is that the dataset accurately represents digital banking behavior and fraud patterns. Another assumption is that the fraud labels included in the dataset were correctly identified.

The analysis also assumes that historical behavioral patterns can help predict future fraudulent activity. Additionally, it is assumed that the selected variables contain meaningful relationships with fraud risk and that customer behavior remains relatively stable throughout the analysis period.

Limitations

One limitation of this project is that the dataset may not fully represent real-world banking environments or continuously evolving fraud techniques. Fraud methods constantly change over

time, meaning that patterns identified in the dataset may not perfectly reflect future fraud behavior.

Another limitation involves class imbalance because fraudulent transactions are much less common than legitimate transactions. This imbalance may affect machine learning performance and create misleading accuracy results if not handled properly.

The dataset may also contain less behavioral or contextual information compared to real production banking systems. Therefore, the findings from this project may not fully generalize across all financial institutions or fraud detection environments.

Challenges

One major challenge in this project involved handling class imbalance within the dataset. Since fraudulent transactions are less common, machine learning models may become biased toward predicting legitimate transactions more frequently.

Another challenge involved preventing machine learning models from overfitting while still maintaining strong predictive performance. More advanced models such as Random Forest and XGBoost improved performance but also increased model complexity and reduced interpretability.

Communicating technical findings clearly to non-technical audiences was also an important challenge. Since fraud detection affects both business operations and customer experience, the findings and recommendations needed to remain understandable from both technical and business perspectives.

Future Uses/Additional Applications

The findings from this project may support additional fraud prevention applications within digital banking systems. Financial institutions may be able to use similar machine learning approaches for:

- real-time fraud monitoring
- adaptive authentication systems
- anomaly detection

- transaction risk scoring
- account takeover prevention

Future research could expand the analysis by incorporating:

- real-time streaming transaction data
- customer segmentation
- geographic analysis
- deep learning techniques
- additional cybersecurity-related variables

As fraud techniques continue to evolve, machine learning models will likely become increasingly important for helping financial institutions detect suspicious activity more efficiently.

Recommendations

Based on the findings from this project, financial institutions should continue strengthening authentication systems and monitoring suspicious behavioral patterns more closely.

Banks may benefit from combining authentication analysis with behavioral analytics to improve fraud detection performance. Variables such as suspicious IP activity, repeated login attempts, transaction velocity, and device risk scores should be monitored carefully because they appear to contribute meaningful predictive information.

Financial institutions should also prioritize explainable machine learning models whenever possible. While more complex models such as XGBoost may improve predictive performance, explainability remains important for understanding model decisions and maintaining customer trust.

Another recommendation is that fraud detection systems should be continuously monitored and updated because fraud techniques constantly evolve over time. Regular retraining and evaluation of machine learning models may improve long-term fraud detection performance.

Implementation Plan

The implementation process for this project began with data preparation and exploratory data analysis. After cleaning the dataset and validating the variables, the data was preprocessed for machine learning modeling.

The next stage involved training multiple machine learning models, including Logistic Regression, Random Forest, and XGBoost. The models were then evaluated using performance metrics such as precision, recall, F1-score, ROC-AUC, and confusion matrices.

After comparing model performance, visualizations and business-focused insights were developed to communicate findings clearly. The final stage of the project focused on recommendations, conclusions, and interpretation of results from both technical and business perspectives.

Ethical Assessment

There are several ethical concerns related to fraud detection systems and behavioral monitoring within banking environments.

Financial institutions collect large amounts of customer behavioral and transactional data, which raises concerns regarding customer privacy and data security. It is important that banks handle customer data responsibly and follow data protection regulations.

Another ethical concern involves false positives, where legitimate customers may incorrectly be flagged as fraudulent. This may negatively affect customer trust, account accessibility, and the overall banking experience.

Machine learning models may also unintentionally introduce bias if certain behavioral patterns become disproportionately associated with fraud risk. Because of this, fairness, transparency, and explainability are important considerations throughout the project.

Financial institutions should regularly monitor and evaluate fraud detection systems to reduce bias, maintain fairness, and ensure responsible use of predictive analytics.

References

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Q & A

1. Why did you choose this topic?

I chose this topic because digital banking fraud has become a major issue as more customers rely on online and mobile banking. Since I currently work in banking, I was interested in understanding how authentication methods and behavioral analytics can help reduce fraud risk while still maintaining a positive customer experience.

2. Why is fraud detection important in banking?

Fraud detection is important because fraudulent transactions can lead to financial losses, cybersecurity risks, and reduced customer trust. Financial institutions need systems that can identify suspicious activity quickly while minimizing inconvenience for legitimate customers.

3. Why did you choose Logistic Regression, Random Forest, and XGBoost?

I selected Logistic Regression because it provides a simple and explainable baseline model. Random Forest and XGBoost were selected because they are more advanced ensemble learning methods that can capture more complex relationships and improve predictive performance.

4. Why was recall important in this project?

Recall was especially important because false negatives in fraud detection can allow fraudulent transactions to go undetected. In banking environments, missing fraudulent activity may create larger financial and security risks compared to incorrectly flagging some legitimate transactions.

5. Which variables were most associated with fraud?

Some of the strongest predictors included transaction velocity score, suspicious IP indicators, device risk score, and repeated login attempts. These variables appeared to have meaningful relationships with fraud risk during the analysis.

6. What challenges did you face during the project?

One major challenge was class imbalance because fraudulent transactions were much less common than legitimate transactions. Another challenge involved balancing predictive performance with model explainability, especially when using more advanced models such as XGBoost.

7. How could banks use your findings in real-world situations?

Banks could use similar machine learning approaches for real-time fraud monitoring, adaptive authentication systems, anomaly detection, and transaction risk scoring. The findings may also help financial institutions strengthen fraud prevention systems while maintaining customer trust.

8. Why is explainability important in banking models?

Explainability is important because financial institutions need to understand why a model makes certain predictions, especially when decisions may impact customer accounts or transactions. More explainable models also help improve transparency and customer trust.

9. What would you improve if you had more time or data?

If I had more time or access to additional data, I would explore real-time transaction monitoring, deep learning methods, geographic analysis, and customer segmentation. I would also like to test the models using larger real-world banking datasets to improve generalizability.

10. Why not just use the model with the highest accuracy?

The model with the highest accuracy is not always the best choice, especially with imbalanced fraud datasets. Accuracy alone can be misleading if the model mainly predicts legitimate transactions. Metrics such as recall, precision, and ROC-AUC provide a better understanding of fraud detection performance.